

LAMPNODE · DESIGN DECISION RECORD

LampNode TRL 2 review decisions

Design Molecule Lab · designmolecule.com

DOCUMENT CONTROL

Document ID	LPN-DDR-001	Version	0.3
Status	Draft	Date	2026-10-02
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/lampnode		
Source	docs/decisions/0001-trl2-review-decisions.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.3	2026-10-02	Amish Chadha	O1 (first partner and region) decided by Amish on 2026-10-02
0.2	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002)
0.1	2026-09-25	Amish Chadha	Record the TRL 2 review recommendations adopted for TRL 3 work under Amish's 2026-09-25 instruction, open for his review, and the items that remain open

0001: TRL 2 review decisions

- **Date:** 2026-09-25
- **Status:** D1 to D10 and O2 to O8 decided by Amish, 2026-09-25: go with recommendation (see LPN-DDR-002). O1 had no recommendation; it was decided by Amish on 2026-10-02 as recommended in LPN-DEC-001.

■ Context

The TRL 2 review note (`docs/REVIEW.md` , session of 2026-09-25) listed eleven LampNode design questions as "Proposed, awaiting Amish", ten of them with a recommendation. On 2026-09-25 Amish asked for this batch of repos to be taken through the usual process with the instruction "you know the drill, nothing gets past TRL 3". He has not reviewed the LampNode items one by one. Under that instruction, every item that carried a recommendation is adopted as recommended for TRL 3, open for his review. Items without a recommendation stay open. Later on 2026-09-25 Amish wrote "i accept all your recommendations, go with them across all repos", so every item below that carries a recommendation is now decided by Amish (LPN-DDR-002). TRL 4 is on hold by his instruction.

■ Options considered

The options for each item are in `docs/REVIEW.md` (TRL 2 session) and LPN-PRC-001 v0.2. They are not repeated here.

■ Decision

Table 1. Items decided by Amish on 2026-09-25.

#	Item	Status and recommendation adopted	Where it now lives
D1	Socket	Decided by Amish, 2026-09-25: go with recommendation. ANSI C136.41 7-contact first, for retrofits; Zhaga Book 18 as a later variant	LPN-PRC-001 v0.3, LPN-REQ-001 R1, <code>cad/src/model.py</code>
D2	Presence sensor	Decided by Amish, 2026-09-25: go with recommendation. 24 GHz Doppler radar, with schedule-only operation kept as a firmware mode	LPN-PRC-001 v0.3, LPN-REQ-001 R6
D3	Sensor head link	Decided by Amish, 2026-09-25: go with recommendation. Cable from the controller to a head clamped under the arm	LPN-PRC-001 v0.3, <code>cad/src/model.py</code> , LPN-DWG-001
D4	Radio	Decided by Amish, 2026-09-25: go with recommendation. LoRaWAN on the STM32WL-class module shared with FieldNode and TwinKit	LPN-PRC-001 v0.3, LPN-REQ-001 R15
D5	Dimming interface	Decided by Amish, 2026-09-25: go with recommendation. 0 to 10 V only; DALI-2 D4i on the auxiliary contacts only if a partner's stock needs it. No partner has been named, so D4i is not in the TRL 3 design	LPN-REQ-001 R4 (redefined)
D6	Fail-on relay	Decided by Amish, 2026-09-25: go with recommendation. Normally closed contact, coil energized in daytime. LPN-CAL-001 puts the average coil power at 0.203 W, not about 0.13 W	LPN-PRC-001 v0.3, LPN-CAL-001 section 3
D7	Hosted sensor power	Decided by Amish, 2026-09-25: go with recommendation. Sibling sensors (AirStreet, NoiseMap) stay on FieldNode solar; LampNode's 12 V ports are an option for them. Hosted sensors that need power around the clock on cabinet-switched feeders bring their own storage. Wording in sibling repos is to be coordinated, not changed here	LPN-REQ-001 R13 (redefined)
D8	Default policy	Decided by Amish, 2026-09-25: go with recommendation. Ships with photocell-equivalent full output; any dimming profile is set by the road owner; the reference profile is for estimates only	LPN-PRC-001 v0.3, LPN-REQ-001 assumptions
D9	Mains range	Decided by Amish, 2026-09-25: go with recommendation. 120 to 277 V now; a 347 V and 480 V variant deferred until a partner needs it	LPN-REQ-001 R2 (redefined)
D10	Interoperability	Decided by Amish, 2026-09-25: go with recommendation. A TALQ bridge is CityTwin scope, not a LampNode requirement	LPN-REQ-001 R15 (redefined)

No budget change, and no reworded pitch or problem line, was recommended at TRL 2, so `budget_usd` (150), the pitch and the problem in `project.yaml` and `README.md` are unchanged.

Items that remain open

Table 2. Items open at v0.1, with their status at v0.2.

#	Item	Status
O1	First partner (city, utility or campus) and region for co-design	No recommendation was made at TRL 2. Decided by Amish, 2026-10-02 (LPN-DEC-001): a US partner on the 915 MHz band; first candidate to approach: a university campus or municipal utility in the Dallas-Fort Worth area that runs LED streetlights with ANSI C136.41 sockets
O2	R7 shortfall. LPN-CAL-001 gives 34.1 to 36.4 % against 35 %. Options: (a) relax R7 to 30 % with the reference profile; (b) keep 35 % and move the floor start to 21:00 in the reference profile; (c) keep both and accept R7 at risk until real traffic counts exist. Recommendation: (c), because the road owner sets the profile and the traffic assumption dominates	Decided by Amish, 2026-09-25: go with recommendation: option (c); R7 stays at 35 % and at risk
O3	R13 hosted power at high temperature (3.90 W peak against 3.55 W at 65 °C). Options: (a) firmware limits the ports to 2.5 W above 50 °C inside; (b) a 10 W supply module (about +\$3, slightly more no-load loss). Recommendation: (a)	Decided by Amish, 2026-09-25: go with recommendation: option (a), firmware port limit; applied in LPN-REQ-001 R13 and LPN-CAL-001 section 11
O4	TCXO real-time clock for R5 (+\$3)	Decided by Amish, 2026-09-25: go with recommendation. In bom/bom.csv line 7
O5	Relay: normally closed with an 80 A or better inrush rating, closed at the voltage zero crossing (+\$2)	Decided by Amish, 2026-09-25: go with recommendation. In bom/bom.csv line 5; the inrush rating per driver model is checked at TRL 4, on hold
O6	Surge stage: 385 V class 20 mm varistor line to neutral and TVS diodes on the low-voltage leads, no gas discharge tube, since the socket has no earth contact (+\$1)	Decided by Amish, 2026-09-25: go with recommendation. In bom/bom.csv line 3
O7	Fail-safe logic: a charge-pump coil drive that drops the relay (lamp on) if the controller stops toggling it, and a day decision that needs both the clock and the light sensor; I/Q radar to reject rain	Decided by Amish, 2026-09-25: go with recommendation. In LPN-PRC-001 and bom/bom.csv lines 7 and 12; firmware beyond a sketch is TRL 4, on hold
O8	Expansion ports change from 4-pole to 5-pole M12 to match FieldNode's two M12 5-pin ports (FND-DDR-001 D5); the pinout itself is open at FieldNode (O2 there)	Decided by Amish, 2026-09-25: go with recommendation. 5-pole ports in bom/bom.csv lines 10 and 14; the pinout is a cross-repo action with FieldNode

■ Consequences

- R2, R4, R13 and R15 are redefined in LPN-REQ-001 v0.3 to match D9, D5, D7 and D10. The TRL 2 "not met" entries for 347 V and 480 V, D4i, daytime hosted power and TALQ become scope notes, not failures.
- The TRL 3 design costs \$130.00 against the unchanged \$150 budget (LPN-CAL-001 section 13).
- R3, R6, R7, R11 and R13 are at risk on paper and are listed in docs/REVIEW.md .
- All items except O1 are decided by Amish (LPN-DDR-002); O1 was decided on 2026-10-02 (LPN-DEC-001). R13 is met on paper with the O3 firmware port limit; R3, R6, R7 and R11 stay at risk.